

Máximo Fernández Núñez

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AI engineer, +13 years shipping production systems. Focused on LLM agents: architecture, tooling, observability and deployment. Background in computer vision and embedded AI on edge hardware.

PROFESSIONAL EXPERIENCE

Unusuals

Remote

AI Agent Engineer

January 2026 – July 2026

- Sole engineer behind the agent of an inspection assistant for Spain's national grid operator (REE), letting field technicians query millions of high-voltage pylon photos by region and defect type instead of reviewing towers one by one. In production, used by real teams.
- Redesigned the architecture from single intent-to-tool routing to a multi-step ReAct loop over ~10 tools (image analysis, S3 navigation, vision-model training, email reporting), turning a command-driven bot into an autonomous agent able to resolve multi-region requests end to end in one conversation.
- Added multi-provider LLM routing (OpenAI, Mistral, OpenRouter) with automatic failover on provider outages, plus step limits, guardrails and prompt caching to reduce token spend.
- Introduced LangSmith tracing: root-cause analysis dropped from 10+ minutes of manual log digging to seconds, and user-reported failures became exactly reproducible, previously they took days or weeks to reproduce.
- Built per-user memory that adapts to individual preferences, slash commands, and Microsoft Teams and web interfaces. Owned deployment and infrastructure.

Freelance

Remote

Generative AI | Agents | RAG | chatbots

September 2024 – January 2026

- Delivered a multi-agent tutoring system (one agent per subject) for Turing Dream, used by secondary-school students in Spain and Colombia, with course material authored and validated by a team of teachers. Agent planning, sub-agents, tool design, context engineering and human-in-the-loop.
- Built a voice agent for kitchen and floor staff at CIVIR to consult and prepare recipes and menus. Designed a streaming pipeline (100 ms audio chunks, VAD-gated buffering, silence-based end-of-speech detection) that sent only real speech to the STT API, sharply cutting transcription cost.
- Owned backend, frontend and deployment of the agent side of the product, working alongside a 3-person team on the SaaS platform and infrastructure. Direct requirements gathering with clients.

Bravent

Remote

AI Tech Lead

Sept 2024 – November 2024

Technical leadership on a generative-AI initiative on Azure: architecture, technology selection and risk assessment. The programme was halted by the client.

Sener Aeroespacial

Madrid, Spain

Machine learning engineer

Sept 2022 – Sept 2024

- Built the perception stack (segmentation, classification, detection, depth estimation) for an autonomous ground vehicle operating in unstructured terrain, within a European defence consortium.
- Optimised inference with TensorRT on NVIDIA AGX Orin: ~30% lower latency and VRAM usage for ~2% accuracy loss, fitting the full stack on embedded hardware.
- Created the segmentation dataset from scratch and fine-tuned models where no public dataset covered the target objects. Built an annotation tool that cut labelling time per image from hours to minutes.
- Supervised a colleague's master's thesis on a RAG system over internal project documentation. Mentored 3 interns and junior engineers, and successfully backed a vision-team colleague's case for promotion.

Arquimea

Machine Learning & Embedded Engineer

Madrid, Spain

October 2017 – July 2022

- Proposed and delivered a single hardware platform with shared low-level firmware across 3 UAV models, eliminating per-model HW/FW redesign and letting features built for one aircraft be inherited by the rest.
- Replaced a commercial object-detection module with an in-house YOLO detector, saving €2,000 per unit.
- Designed schematics and PCBs and programmed the microcontrollers for UAV flight control. One resulting system is deployed at several airports for runway bird dispersal.
- Prototyped GPS-denied geolocation: trained a network on overhead imagery of Spain and France that localised a UAV from a single nadir image to ~1 km error, enough to return an aircraft to friendly territory when GPS is jammed, instead of losing it behind enemy lines. MVP, not taken to production.

Indra

Electronic engineer

Madrid, Spain

January 2013 – Oct 2017

- Designed control PCBs and firmware for an autonomous helicopter programme for the Spanish Ministry of Defence.

EDUCATION

Universidad Complutense de Madrid

Madrid, Spain

Electronic engineering

June 2011

Honors in the subjects of Systems control, Introduction to Semiconductors, Robotics, Introduction to Semiconductors, Computer architecture and in Final degree project.

PROJECTS & COMMUNITY

- Gymnasia, github.com/maximofn/gymnasia, Fitness app running two agents fully on-device, with no backend: user data never leaves the phone. A BYOK conversational coach (adapters for OpenAI, Anthropic and Google, 12 local tools, streaming tool-use loop, remote system prompt with offline fallback) and a vision sub-agent that estimates macros from meal photos, with barcode lookup against OpenFoodFacts. Architecture documented at gymnasia-sable.vercel.app.
- "Welcome to la secta": Weekly AI podcast, 800 subscribers, ~500 views per episode, with guests from the Spanish-speaking AI scene. Built a generative-AI pipeline that renders, uploads, clips and distributes each episode across platforms automatically, including voice cloning and lip-sync for future multilingual editions.
- maximofn.com: AI blog and technical writing, built and self-operated: Astro on Vercel behind Cloudflare as reverse proxy, WAF and media CDN. ~2,000 LinkedIn followers, posting near-daily.

SKILLS

- Agents & LLMs: LangGraph, LangChain, LangSmith, MCP, OpenAI Agents SDK, ReAct architectures, RAG, vector databases, guardrails, prompt caching, multi-provider failover
- ML & vision: PyTorch, TensorRT, HuggingFace, YOLO, segmentation, detection, depth estimation, fine-tuning, dataset creation
- Engineering: Python, SQL, Docker, git, CI/CD, testing, code review, bash/SSH, Gradio, Astro, HTML/CSS
- Languages: Spanish (native), English (full professional proficiency), Portuguese (elementary)